



Maintenance Schedule

Diesel Engine
EU Stage IIIB
12V1600Rx0
12V1600Rx0L
Application group 2A

MS50137/04E

Engine model	Power per cylinder	Application group
12V1600R70 (EU IIIB with SCR system)	47 kW/cyl.	2A, continuous operation, unrestricted
12V1600R70L (EU IIIB with SCR system)	52 kW/cyl.	2A, continuous operation, unrestricted
12V1600R80 (EU IIIB with SCR system)	55 kW/cyl.	2A, continuous operation, unrestricted
12V1600R80L (EU IIIB with SCR system)	58.3 kW/cyl.	2A, continuous operation, unrestricted

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

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Table of Contents

1 Maintenance Schedule		
1.1 Preface	4	
1.2 Special conditions	5	
1.3 Maintenance task tolerances	6	
1.4 Allocation matrix application group 2A	7	
1.5 Noncyclical maintenance tasks	8	
		1.6 Cyclical maintenance tasks 42000 HRS 9
		1.7 Cyclical maintenance tasks 36000 HRS 11
		1.8 Cyclical maintenance tasks 30000 HRS 13
		1.9 Cyclical maintenance tasks 27000 HRS 15
		1.10 Cyclical maintenance tasks 24000 HRS 17

1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product manufactured by Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of operational availability. The maintenance intervals as well as the required maintenance task scopes are based on operational experience and are therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of demanding operational and ambient conditions.

Observance of the specified maintenance intervals is essential to maintain product safety.

The intervals at which the maintenance tasks are to be performed are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule)

QL3: Maintenance work which requires partial disassembly of the product

QL4: Maintenance work which requires complete disassembly of the product

Additional notes on maintenance and preservation:

Maintenance activities stipulated in any documentation provided by third-parties (suppliers and component manufacturers), particularly those relating to individual components and subsystems, may differ from the information specified in this Maintenance Schedule. The Maintenance Schedule provided by Rolls-Royce Solutions shall take precedence in such case.

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

Refer to the applicable Fluids and Lubricants Specifications for details of fluid and lubricant change intervals.

If the product is to remain out of service for a longer period of time, Rolls-Royce Solutions recommends carrying out a monthly test run until the engine has reached its normal operating temperature. If the engine is scheduled to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications.

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

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1.2 Special conditions

Note

The engines are certified for operation with approved fuels as listed in the Fluids and Lubricants Specifications.

The maintenance intervals for the components specified in the maintenance schedule refer to engine operation with diesel fuel with a sulfur content < 15 ppm, e.g. EN 590.

If fuels with a sulfur content > 15 ppm are used, the maintenance intervals specified in the maintenance schedule for the affected components could be shorter.

Engines with exhaust gas recirculation and/or exhaust gas aftertreatment must not be operated with excessive sulfur content in the fuel. The limit values in the Fluids and Lubricants Specifications apply.

Table 2: Special conditions

1.3 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.4 Allocation matrix application group 2A

Load profile		Load factor %	Load indicator %	0 to 47 kW/cyl.	48 to 55 kW/cyl.	56 to 58 kW/cyl.	58.3 kW/cyl.		
Load %	Time %								
110 ₍₁₀₀₋₁₁₀₎	0	≤ 40	≤ 12	42,000	36,000	30,000	27,000	-	Standard
100 ₍₉₀₋₁₀₀₎	8								
90 ₍₈₀₋₉₀₎	7								
80 ₍₆₅₋₈₀₎	2								
65 ₍₅₀₋₆₅₎	2								
50 ₍₃₀₋₅₀₎	3								
30 ₍₅₋₃₀₎	70								
Idle Speed ₍₀₋₅₎	8	≤ 50	≤ 30	36,000	30,000	27,000	24,000	-	Load Profile
110 ₍₁₀₀₋₁₁₀₎	0								
100 ₍₉₀₋₁₀₀₎	27								
90 ₍₈₀₋₉₀₎	4								
80 ₍₆₅₋₈₀₎	8								
65 ₍₅₀₋₆₅₎	4								
50 ₍₃₀₋₅₀₎	8								
30 ₍₅₋₃₀₎	17								
Idle Speed ₍₀₋₅₎	32								

1.5 Noncyclical maintenance tasks

No.	Qualification	Task	Maintenance tasks	Reference task	Offset	Value	Unit	Option
0001	QL1	WM00582	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours.		0	1000	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

1.6 Cyclical maintenance tasks 42000 HRS

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check engine oil level		WM00285
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Visually inspect engine for leaks and check its general condition		WM00286
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	When engine is running, check for abnormal running noises, exhaust gas color and vibrations		WM00289
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Drain water and contaminants from fuel prefilter		WM00290
QL1	-	1 DAY	AIR SUPPLY SYSTEM	Check service indicator of air filter, replace air filter if necessary	X	WM02056
QL1	100	18 YR	AIR PIPING TO CYLINDERS	Clean intercooler and check for leaks or replace	X	WM00986
QL1	1000	1 YR	LUBE OIL SYSTEM	Replace oil filter at every engine oil change, or when the time limit (years) is reached, at the latest		WM00067
QL1	1000	1 YR	AUXILIARY POWER TAKE-OFF	Check belt condition and tension, replace belt if necessary		WM00547
QL1	1000	2 YR	FUEL SYSTEM (LOW PRESSURE)	Fuel prefilter, replace filter		WM00084
QL1	1000	2 YR	FUEL SYSTEM (LOW PRESSURE)	Replace fuel filter or fuel filter element		WM00583
QL1	2000	2 YR	RUBBER BELLOWS	Check all elastomer bellows in the coolant, lube oil, combustion air, fuel and charge-air system, replace if necessary		WM00552
QL1	2000	2 YR	HOSE	Check all hose lines in the coolant, lube oil, combustion air, fuel and charge-air system, replace if necessary		WM00553
QL1	2000	2 YR	AIR SUPPLY SYSTEM	Replace air filter in accordance with maintenance intervals. Alternatively: Perform air filter replacement as per intake-air measurement and control message	X	WM01260
QL1	2000	9 YR	VALVE GEAR	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours		WM00582
QL1	6000	2 YR	AUXILIARY POWER TAKE-OFF	Replace drive belt of engine coolant pump		WM00557
QL1	12000	9 YR	CRANKCASE VENTILATION	Replace oil separator (impactor)		WM00106
QL1	15000	9 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace fuel injection valve/injector		WM00090
QL1	15000	9 YR	GOVERNOR	Enter injector coding (IIG)		WM00752
QL1	42000	9 YR	PRESSURE PIPE NECK	Replace pressure pipe neck in cylinder head		WM00186
QL1	42000	18 YR	HP FUEL LINE HP LINE	Replace HP line		WM00172
QL3	21000	6 YR	RUBBER BELLOWS	Replace all elastomer bellows in coolant, lube oil, combustion air, fuel and charge-air system		WM00558
QL3	21000	6 YR	HOSE	Replace all hose lines in coolant, lube oil, combustion air, fuel and charge-air system		WM00559
QL3	21000	9 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace HP pump		WM00091
QL3	21000	9 YR	EXHAUST TURBOCHARGER	Replace exhaust turbocharger		WM00117
QL3	21000	9 YR	ENGINE MOUNTING	Replace rubber element of resilient mounts		WM00127
QL3	21000	9 YR	COOLANT SYSTEM	Replace actuator		WM00791
QL3	21000	9 YR	STARTING SYSTEM	Overhaul or replace starter	X	WM00971
QL3	21000	9 YR	WINDOW GASKET	Fit new window gasket		WM01618
QL3	21000	9 YR	RUNNING GEAR	Replace gaskets on vibration damper		WM01693
QL3	42000	18 YR	EXHAUST SYSTEM	Replace exhaust pipe bellows	X	WM00072
QL3	42000	18 YR	SENSOR TECHNOLOGY AND WIRING HARNESS	Replace wiring harness		WM00137
QL3	42000	18 YR	COOLANT SYSTEM	Replace engine coolant pump		WM00762
DAY = Days HRS = Hours YR = Years						

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